

End-to-End System: Sensor → Inference → BLE → Cabin Control

**Wear OS
Sensors
PPG · EDA
Temp · ACC**

```
graph LR; A[Wear OS Sensors  
PPG · EDA  
Temp · ACC] --> B[TFLite Stress Classifier  
[30 × 5]]; B --> C[BLE Transmit  
12-byte packet]; C --> D[Arduino Cabin Controller  
FSM]; D --> E[Adaptive Environment  
Light · Fan  
Audio];
```

The diagram illustrates a five-stage end-to-end system. It begins with a green box for 'Wear OS Sensors' (PPG, EDA, Temp, ACC), followed by a blue box for 'TFLite Stress Classifier' (30 x 5). The third stage is an orange box for 'BLE Transmit' (12-byte packet). This is followed by a pink box for 'Arduino Cabin Controller' (FSM), and finally a purple box for 'Adaptive Environment' (Light, Fan, Audio). Arrows indicate the sequential flow from left to right between each stage.

**TFLite
Stress
Classifier
[30 × 5]**

**BLE
Transmit
12-byte
packet**

**Arduino
Cabin
Controller
FSM**

**Adaptive
Environment
Light · Fan
Audio**